Qualification Exam Notes

Pablo Bustillo Vazquez

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1 Syllabus

The topic of this Qualification Examination is the computation of the classifying space of the ∞-category of cobordism, from the seminal work of Galatius, Madsen, Tillman and Weiss [GMTW09]. We state here their main result. Fix $d \geq 0$ an integer.

Definition 1.1. Let $\mathcal{X}$ be the $(1,1)$-category of manifolds without boundary and smooth maps, together with its usual Grothendieck topology.

Definition 1.2. We have a functor $|-|\colon \mathrm{Sh}(\mathcal{X}) \to \mathcal{S}$ given by

$$\mathcal{F} \mapsto \varinjlim_{[n]\in\Delta^{\mathsf{op}}} \mathcal{F}(\Delta^n_e)$$

A morphism of sheaves is called a **weak homotopy equivalence** if the induced map on spaces is a weak homotopy equivalence.

For $a, b\colon X \to \mathbb{R}$, we write $a < b$ when $\forall x \in X, a_0(x) \leq a_1(x)$ and $\{x \in X \mid a_0(x) = a_1(x)\}$ is clopen. We also write $X \underline{\times} (a,b) = \{(x,t) \in X \times \mathbb{R} \mid a(x) < t < b(x)\}$.

Definition 1.3. We have a $\mathrm{Sh}(\mathcal{X})$-internal category $\mathcal{C}^\perp$ whose:

- collection of objects is the sheaf

$$\mathsf{ob}(\mathcal{C}^\perp)(X) = \{a\colon X \to \mathbb{R}, M \subset_d X \times \mathbb{R}^\infty \twoheadrightarrow X \text{ proper submersion with } d\text{-dimensional fibers}\}$$

- collection of morphisms is the sheaf

$$\mathsf{mor}(\mathcal{C}^\perp)(X) = \varinjlim_{\epsilon\colon X\to\mathbb{R}} \bigsqcup_{a_0<a_1\colon X\to\mathbb{R}} \mathcal{C}^\perp(X, a_0, a_1, \epsilon)$$

  where $\mathcal{C}^\perp(X, a_0, a_1, \epsilon)$ is the set of $W \subset X \underline{\times} (a_0 - \epsilon, a_1 + \epsilon) \times \mathbb{R}^\infty$ such that:

  (a) (smooth family) $W \twoheadrightarrow X$ is a submersion with d-dimensional fibers,

  (b) (compact cobordisms) $W \to X \underline{\times} (a_0 - \epsilon, a_1 + \epsilon)$ is proper,

  (c) (tubularity) corestricted to $X \underline{\times} (a_i - \epsilon, a_i + \epsilon)$ for $i \in \{0,1\}$, W splits as a product.

This gives a category object in spaces after applying the functor $|-|$. The latter is the object of interest here, which we call the ∞-category of cobordism.

Theorem 1.1 (Goal)

The classifying space of the ∞-category of cobordism is

$$\Omega^{\infty-1} MTO(d) = \varinjlim_{n\to\infty} \Omega^{n+d-1} \mathsf{Thom}(\gamma_d^\perp \to \mathsf{Gr}_d(\mathbb{R}^{n+d}))$$

The core of the proof lies in using Phillips Submersion Theorem, one of the first instance of a **microflexible h-Principle**. Instead of following Phillips' original proof to the letter, we will follow an exposition of his proof by Haeflinger [Hae71], further elucidated by Geiges [Gei03]. At no cost, one can change notations to transform this argument into a proof of a microflexible h-Principle for general isotopy-invariant $\mathcal{X}$-sheaves, which we will do.

2 Smoothly Parametrized Spaces

2.1 Definitions

Definition 2.1. Let $\mathcal{X}$ be the $(1,1)$-category of manifolds without boundary and smooth maps, together with its usual Grothendieck topology.

Definition 2.2. Let Y be a compact manifold with corners and $\mathcal{F}\colon \mathrm{Sh}(\mathcal{X})$ a sheaf. We define the stalk $\mathcal{F}(Y) = \varinjlim_{Y\subset U} \mathcal{F}(U)$ as the colimit on all fatenings of Y. By abuse of notation, we will denote an element of $\mathcal{F}(Y)$ by $Y \to \mathcal{F}$.

Note that $\mathcal{F} \mapsto \mathcal{F}(Y)$ commutes with finite limits but not all limits. It's not a corepresentable sheaf, but as long as we never consider infinite limit diagrams, we have some flexibility with this notation.

Remark 2.1. Note that compact manifolds with corners are compact objects in $\mathrm{Sh}(\mathcal{X})$.

Definition 2.3. We have a functor $|-|\colon \mathrm{Sh}(\mathcal{X}) \to \mathbf{Top}$ given by

$$\mathcal{F} \mapsto |[n] \mapsto \mathcal{F}(\Delta_e^n)|$$

A morphism of sheaves is an **equivalence** if the induced map on spaces is a weak homotopy equivalence.

Definition 2.4. A morphism of sheaves $\mathcal{F} \to \mathcal{G}$ is called a **microfibration** if for all X compact manifold (with or without boundary), there exists $0 < \epsilon \leq 1$ and a lift

$$\begin{tikzcd}
 & X \times \{0\} \arrow[r] \arrow[dl] \arrow[d] & \mathcal{F} \arrow[d] \\
X \times [0,\epsilon] \arrow[r] \arrow[ur, dashed] & X \times [0,1] \arrow[r] & \mathcal{G}
\end{tikzcd}$$

It's called a **fibration** (concordance lifting property) if one can always pick $\epsilon = 1$.

Remark 2.2. Note that the pullback of a fibration is a fibration.

2.2 Concordances

Definition 2.5. Let X be a smooth manifold and $\mathcal{F}\colon \mathrm{Sh}(\mathcal{X})$ be a sheaf. There is a natural equivalence relation on $\mathcal{F}(X)$ given by $u \sim v$ when we have

$$\begin{tikzcd}
X \times \{0\} \arrow[r] \arrow[dr, "u"'] & X \times [0,1] \arrow[d, dashed] & X \times \{1\} \arrow[l] \arrow[dl, "v"] \\
 & \mathcal{F} &
\end{tikzcd}$$

The set of equivalence classes is denoted $\mathcal{F}[X]$. Let $A \subset X$ be a closed subset and $s \in \varinjlim_{A\subset U} \mathcal{F}(U)$ be a germ. We define the set $\mathcal{F}[X, A, s]$ to be the set of sections which restrict to s around A, up to the equivalence $u \sim v$ above given by these $X \times [0,1] \to \mathcal{F}$ factoring through s in some neighbourhood of A.

Proposition 2.1 ([MW07, Prop. 2.17, A.1])

Let $X\colon \mathcal{X}$, $A \subset X$ closed and $z \in \mathcal{F}(\bullet)$. We have $\mathcal{F}[X, A, z] \simeq \pi_0[(X, A), (|\mathcal{F}|, z)]$ natural in $\mathcal{F}$.

Remark 2.3 (Extended Triangulation). For any triangulation of X with vertex set $\mathbb{T}$, one can construct the following data, called an **extended triangulation** of X. For all $S \in \mathbb{T}$ distinguished set (set of vertices of a simplex), we denote $c_S\colon \Delta(S) \hookrightarrow X$ the embedding and $\Delta_e(S)$ the extended simplex. We can pick embeddings $(c_{S,e}\colon \Delta_e(S) \hookrightarrow X)_{S\subset\mathbb{T} \text{ distinguished}}$ extending the c_S's and $h\colon X \times [0,1] \to X$ such that:

- $h_t = 1_X$ for t in a neighbourhood of 0,
- h_t constant for t in a neighbourhood of 1,
- for all t, $h_t(c_S(\Delta(S))) \subset c_S(\Delta(S))$,
- for all t, $h_t(c_{S,e}(\Delta_e(S))) \subset c_{S,e}(\Delta_e(S))$,
- for all $S \in \mathbb{T}$ distinguished set, we have an open neighbourhood $\Delta(S) \subset V_S$ such that $h_1(V_S) \subset c_S(\Delta(S))$.

Pablo: Draw illustration on the Kindle

Proof: First we prove the result for $A = \emptyset$.

- **Defining** $\xi\colon \mathcal{F}[X] \to \pi_0[X, |\mathcal{F}|]$ Pick an extended triangulation of X with vertex set $\mathbb{T}$. Let $u \in \mathcal{F}(X)$ be a section. For each $S \in \mathbb{T}$ distinguished set, we have an element $u_S = c_{S,e}^* u \in \mathcal{F}(\Delta_e(S))$ of $|\mathcal{F}|(S)$. This gives a unique map $\xi(u)\colon X \to |\mathcal{F}|$ such that each of the following diagrams commute

$$\begin{tikzcd} \Delta(S) \arrow[d, "c_S"'] \arrow[dr, "u_S"] & \\ X \arrow[r, "\xi(u)"'] & |\mathcal{F}| \end{tikzcd}$$

  as X is the colimit (and homotopy colimit) of the simplices of its triangulation. It's clear that this only depends on the equivalence class of u, as changing it corresponds to having a homotopy of cocone diagrams. It's also clear that this is stable under refinements. **Importantly**, this is natural in $\mathcal{F}$.
- **Defining** $\nu\colon \pi_0[X, |\mathcal{F}|] \to \mathcal{F}[X]$ Let $g\colon X \to |\mathcal{F}|$. By the Simplicial Approximation Theorem, can find a triangulation of X such that g is homotopy equivalent to a simplicial map, and we can extend this triangulation. To each $S \in \mathbb{T}$ distinguished set is associated an element $g \circ c_S \in |\mathcal{F}|(\Delta(S)) = \mathcal{F}(\Delta_e(S))$. We can define a section $\nu(g) \in \mathcal{F}(X)$ by glueing the sections

$$V_R \xrightarrow{h_1} c_S(\Delta(S)) \hookrightarrow c_{S,e}(\Delta_e(S)) \xrightarrow{c_{S,e}^{-1}} \Delta_e(S) \xrightarrow{g \circ c_S} \mathcal{F}$$

  It's clear they do glue, and that this is independent of both the homotopy class of g and the triangulation.

Pablo: practically speaking, this makes the sections constant on a neighbourhood of the skeleton

- **Showing** $\nu \circ \xi = 1$**:** Using the homotopy h, we have a concordance between u and $\nu(\xi(u))$.
- **Showing** $\xi \circ \nu = 1$**:** Let $g\colon X \to |\mathcal{F}|$ and consider the construction above of $\nu(g)$. Define the smooth $H\colon X \times \mathbb{R} \to X$ by

$$H(x,t) = \begin{cases} h_0(x) & \text{if } t \leq 0, \\ h_t(x) & \text{if } t \in [0,1], \\ h_1(x) & \text{if } t \geq 1. \end{cases}$$

  Note we have two maps of sheaves $\pi_0, \pi_1\colon \mathcal{F}(- \times \mathbb{R}) \to \mathcal{F}$ which are homotopic after applying $|-|$. This gives a unique map $G\colon X \to |\mathcal{F}(- \times \mathbb{R})|$ such that each of the following diagrams commute

$$\begin{tikzcd} \Delta(S) \arrow[d, "c_S"'] \arrow[dr, "G_S"] & \\ X \arrow[r, "G"'] & |\mathcal{F}(- \times \mathbb{R})| \end{tikzcd}$$

  where G_S is the composition of the maps

$$\Delta_e(S) \times \mathbb{R} \xrightarrow{(c_{S,e}, 1)} c_{S,e}(\Delta_e(S)) \times \mathbb{R} \xrightarrow{H} c_{S,e}(\Delta_e(S)) \xrightarrow{c_{S,e}^{-1}} \Delta_e(S) \xrightarrow{g \circ c_S} \mathcal{F}$$

  Clearly, $g = |\pi_0 \circ G| \simeq |\pi_1 \circ G| = \xi \circ \nu(g)$.

Now the case where $A \neq \emptyset$, we can repeat the same argument, by considering only triangulations where the simplex which meet A are contained in the neighbourhood where the section is constant at z. ■

Corollary 2.2 ([Clo23, Thm. E])

$|-|$ sends filtered colimits to homotopy colimits.

Proof: We simply need to check $\varinjlim \mathcal{F}[X, A, z] \simeq (\varinjlim \mathcal{F})[X, A, z]$, which is a direct conseuquence of Remark 2.1. ■

Corollary 2.3 ([MW07, Prop. A.6])

$|-|$ sends a pullback square by a fibration to a homotopy pullback.

Proof: Consider

$$\begin{tikzcd} |\mathcal{A}_b| \arrow[rr] \arrow[dr] \arrow[dd] & & |\mathcal{C}_d| \arrow[dr] \arrow[dd] & \\ & |\mathcal{A}| \arrow[rr] \arrow[dd] & & |\mathcal{C}| \arrow[dd] \\ \bullet \arrow[rr] \arrow[dr, "b"'] & & \bullet \arrow[dr, "d"'] & \\ & |\mathcal{B}| \arrow[rr] & & |\mathcal{D}| \end{tikzcd}$$

It suffices to show that for any $b \in \mathcal{B}(\bullet)$, the sheaves $\mathcal{A}_b$ and $\mathcal{C}_d$ are homotopy fibers, and that the map $\mathcal{A}_b \to \mathcal{C}_d$ is an equivalence.

- **Homotopy Fibers:** We have a canonical map $|\mathcal{A}_b| \to \mathsf{fib}(|\mathcal{A}| \to |\mathcal{B}|)$, which we want to prove is an equivalence. Let $a \in \mathcal{A}_b(\bullet)$ and $n \geq 0$. One can build an inverse $\pi_n(\mathsf{fib}(|\mathcal{A}| \to |\mathcal{B}|, a) \to \pi_n(|\mathcal{A}_b|, a)$ by considering

$$\begin{tikzcd}
\mathcal{A}_b \arrow[r] \arrow[d] & \mathcal{A} \arrow[d, "\phi"] & S^n \times \{0\} \arrow[l] \arrow[d] \\
\bullet \arrow[r, "b"'] & \mathcal{B} & S^n \times [0,1] \arrow[l] \\
& S^n \times \{1\} \arrow[ul] \arrow[ur] \arrow[uu, dashed] \arrow[uuur, dashed] &
\end{tikzcd}$$

  Left to the reader: Check that this is indeed an inverse.
- **Equivalence:** The map of sheaves $\mathcal{A}_b \to \mathcal{C}_d$ is an isomorphism. ■

Lemma 2.4 (Criterion 2.5)

A map $f\colon \mathcal{F} \to \mathcal{G}$ of sheaves is a weak homotopy equivalence whenever: for all $X\colon \mathcal{X}$, $A \subset X$ closed and $z \in \mathcal{F}(A)$, we have a surjection

$$\mathcal{F}[X, A, z] \twoheadrightarrow \mathcal{G}[X, A, z]$$

Proof: By the proposition above, we have a surjective map $\pi_n(|\mathcal{F}|, z) \to \pi_n(|\mathcal{F}|, f(z))$ for all $n \geq 0$ and based point $z \in \mathcal{F}(\bullet)$. We need to show the kernels are trivial (for $n = 0$, this is to be interpreted in the category of pointed sets). ■

Pablo: to be completed

2.3 Classifying Spaces

Let $\mathcal{A}$ be a $\mathrm{Sh}(\mathcal{X})$-enriched category. We construct here a sheaf $\beta\mathcal{A}\colon \mathrm{Sh}(\mathcal{X})$ together with a map $\mathcal{A} \to \beta\mathcal{A}$ which is a weak homotopy equivalence on classifying spaces.

Definition 2.6. We let

3 Microflexible h-Principle

Definition 3.1. Let $\mathcal{M}\mathsf{fld}_n$ be the $\mathrm{Sh}(\mathcal{X})$-enriched category of finatary n-dimensional manifolds without boundary (interior of compact manifolds with boundary) and open embeddings. More precisely, for $X\colon \mathcal{X}$, we let

$$\mathsf{Emb}(M, N)(X) = \{f\colon X \times M \to X \times N \text{ smooth over } X \mid \forall x \in X, f(x, -) \text{ is an open embedding}\}$$

We consider once again the usual Grothendieck topology on $\mathcal{M}\mathsf{fld}_n$.

Definition 3.2. A sheaf $\Phi\colon \mathcal{M}\mathsf{fld}_n^{\mathsf{op}} \to \mathrm{Sh}(\mathcal{X})$ is called a **microflexible** in M if for all $K_0 \subset K_1 \subset M$ compact and all X compact manifold (with or without boundary),

$$\begin{tikzcd}
& X \times \{0\} \arrow[r] \arrow[dl] \arrow[d] & \Phi_M(K_0) = \varinjlim_{K_0 \subset U} \Phi(U) \arrow[d] \\
X \times [0, \epsilon] \arrow[r] & X \times [0,1] \arrow[r] \arrow[ur, dashed] & \Phi_M(K_1) = \varinjlim_{K_1 \subset U} \Phi(U)
\end{tikzcd}$$

we have a lift. The sheaf is called **flexible** if one can always pick $\epsilon = 1$.

Lemma 3.1

For all sheaves Φ, the stalk $\Phi(\mathbb{R}^n) \simeq \Phi_0$ is an equivalence.

Proof: The maps $\Phi(B_\epsilon(0)) \to \Phi(B'_\epsilon(0))$ are all equivalences, as they clearly come with a homotopy inverse. The colimit indexed by ϵ is a homotopy colimit. Hence the transfinite composition of these maps is an equivalence. ■

3.1 Honmotopy Sheaf Formulation

Lemma 3.2

There is an equivalence of categories between homotopy sheaves and spaces over $B\mathcal{O}(n)$.

Theorem 3.3 (Flexible h-Principle)

If a sheaf Φ is flexible, then $|\Phi|$ is a homotopy sheaf.

If Φ is microflexible in M, it's clearly microflexible in any M' which can be openly embedded in M.

Theorem 3.4 (Microflexible h-Principle)

*If a sheaf Φ is microflexible in M **open**, then $|\Phi|$ is a homotopy sheaf in M.*

3.2 Scanning Map Formulation

Theorem 3.5 (Microflexible Approximation h-Principle)

Let $f\colon \Phi \to \Xi$ be a map between microflexible sheaves on M. Then the following are equivalent:

- *f is a pointwise equivalence,*
- *$f_{\mathbb{R}^n}$ is an equivalence,*
- *f_0 is an equivalence,*

3.3 Application of the h-Principle

3.3.1 Submanifolds

Definition 3.3. Fix $d \geq 0$. Let M^{n+d} be a manifold. Define the sheaf

$$\Phi(M)(X) = \{W \subset_d X \times M \twoheadrightarrow X \text{ of dimension } d \text{ fibers}\}$$

We have for each point $m \in M$

$$\begin{tikzcd}
\gamma_d^\perp \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \gamma_{d,M}^\perp \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \gamma_n^\infty \arrow[d] \\
\mathsf{Gr}_d(\mathbb{R}^{n+d}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathsf{Gr}_d(TM) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathsf{Fl}(n,d) \arrow[d] \\
\bullet \arrow[r, "m"] & M^{n+d} \arrow[r] & \mathsf{Gr}_{n+d}(\mathbb{R}^\infty)
\end{tikzcd}$$

and we define $\mathsf{Thom}^{\mathsf{fib}}(\gamma_{b,M}^\perp) \to M$ to be the fiberwise Thom construction. The assignmenet $M \mapsto \Gamma(\mathsf{Thom}^{\mathsf{fib}}(\gamma_{b,M}^\perp) \to M)$ is a homotopy sheaf and it corresponds to the $\mathcal{O}(n)$-space given by the rotation action on $\mathsf{Thom}(\gamma_d^\perp \to \mathsf{Gr}_d(\mathbb{R}^{n+d}))$.

Lemma 3.6

We have an $\mathcal{O}(n)$-equivariant weak homotopy equivalence

$$|\Phi(\mathbb{R}^n)| \to \mathsf{Thom}(\gamma_d^\perp \to \mathsf{Gr}_d(\mathbb{R}^{n+d}))$$

Proof: Take the "closest point on W to 0" map. It's clearly $\mathcal{O}(n)$-equivariant. One can show that $\Phi(\mathbb{R}^n)[X, A, s] \to \mathsf{Thom}(\gamma_d^\perp \to \mathsf{Gr}_d(\mathbb{R}^{n+d}))[X, A, s]$ is a surjection for all $X \in \mathcal{X}$, $A \subset X$ closed and $s \in \Phi(\mathbb{R}^n)(A)$. Around the closed set A, one just need an isotopy of $\mathbb{R}^n$ that dilates in the direction of the tangent space to the closest point. ■

Corollary 3.7 (of the h-Principle)

We have a natural weak homotopy equivalence

$$|\Phi(M)| \simeq \Gamma(\mathsf{Thom}^{\mathsf{fib}}(\gamma_b^M) \to M)$$

Definition 3.4. Fix M. We define a sheaf of "compactly-supported sections in the M-direction" as follows:

$$\Phi^c(R \times M)(X) = \{W \subset_d X \times \mathbb{R} \times M \twoheadrightarrow X \mid W \to X \times \mathbb{R} \text{ proper}\}$$

Note that we can see this functor as the colimit on all "compactly-thickened paths" in M. This colimit is filtered, hence a homotopy colimit. Restricting to the empty manifold outside that thickened path is taking a homotopy fiber (pullback of a fibration on a point). Hence we get the following result:

Corollary 3.8 (of the h-Principle)

We have a natural weak homotopy equivalence

$$|\Phi^c(\mathbb{R} \times M)| \simeq \Gamma^c(\mathsf{Thom}^{\mathsf{fib}}(\gamma^{\perp}_{b,M\times\mathbb{R}}) \to M)$$

Note that for $M = \mathbb{R}^{n+d-1}$, we have a canonical equivariant

$$|\Phi^c(\mathbb{R} \times \mathbb{R}^{n+d-1})| \simeq \Omega^{n+d-1}\mathsf{Thom}(\gamma^{\perp}_b)$$

as the fiberwise Thom-bundle is then trivial.

3.3.2 Submersions

Proposition 3.9

Let N be a manifold. The sheaf

$$\mathsf{Sub}(M,N)(X) = \{f\colon X \times M \to X \times N \text{ fiberwise submersion over } X\}$$

is microflexible on all manifolds and the sheaf

$$\mathsf{Max}(TM,TN)(X) = \{f\colon X \times TM \to X \times TN \text{ vector bundle map, fiberwise surjective over } X\}$$

is flexible on all manifolds.

We have an obvious $d\colon \mathsf{Sub}(-,N) \simeq \mathsf{Max}(T-,TN)$ given by the fiberwise derivative.

Proposition 3.10

The stalk of the derivative $d_0\colon \mathsf{Sub}(-,N)_0 \simeq \mathsf{Max}(T-,TN)_0$ is an equivalence.

Proof: Pick a partial exponential map $TN \supset U \xrightarrow{e} N$. We construct a map of sheaves $\nu\colon \mathsf{Max}(T-,TN)_0 \to \mathsf{Sub}(-,N)_0$ and show it's a homotopy inverse.

- **Defining the Map:** Consider $X \underline{\times} TB_\epsilon(\mathbb{R}^n) \xrightarrow{f} X \times TN$ for some $\epsilon\colon X \to (0,\infty)$ small enough that $B_\epsilon(T_0\mathbb{R}^n)$ is mapped to U. We define $\nu(f)\colon X \underline{\times} B_\epsilon(T_0\mathbb{R}^n) \to X \underline{\times} TB_\epsilon(\mathbb{R}^n) \to X \times U \xrightarrow{e} N$.
- **Homotopy $d_0 \circ \nu \simeq 1$:** Trivial.
- **Homotopy $\nu \circ d_0 \simeq 1$:** For small enough $\epsilon\colon X \to (0,\infty)$, we can define

$$H(x,m,t) = f(x,0) + \frac{f(x,tm) - f(x,0)}{t}$$

■

3.4 Proof of the h-Principle

Lemma 3.11

Every open finatary manifold M has a finite handlebody decomposition without n-handles.

Proof: Without loss of generality, we can assume M is connected. We can see M as a cobordism from N to $\emptyset$ for N some compact $(n-1)$-manifold. It's equivalent to show that N has a finite handlebody decomposition without 0-handles. Pick a finite handlebody decomposition of M with a minimal amount k of 0-handles. Suppose for the sake of contradiction that $k > 0$. We can see the handlebody decomposition as $N \times [0,1) \bigsqcup \sqcup_{i=1}^k D_i^n$ where $D_i^n \simeq D^n$ on which is attached (in some order) all i-handles for $i > 0$. The only handle attachments that change π_0 are the 1-handles. There must be a step where the connected component of D_0^n gets attached by a 1-handle to one of the other connected components. We can pick a smooth isotopy that slides that attaching map inside D_0^n. We can now remove D_0^n and this 1-handle to the decomposition. This contradicts the minimality of k. ■

Proposition 3.12 ([Gei03, Prop. 3.14] and [Hae71, Prop. 3])

Suppose Φ is microflexible in an open neighbourhood of k-handle $D^k_2 \times D^{n-k}$ for $k < n$. Then the restriction map

$$\Phi(D^k_2 \times D^{n-k}) \to \Phi(D^k_{[1,2]} \times D^{n-k})$$

is a fibration (slight abuse of notations: the context manifold is omitted).

Proof: Let X be a compact parametrizing manifold (with or without boundary) and let us pick

$$\begin{tikzcd} X \times \{0\} \arrow[r, "g_0"] \arrow[d] & \Phi(D^k_2 \times D^{n-k}) \arrow[d] \\ X \times [0,1] \arrow[r, "f"'] & \Phi(D^k_{[1,2]} \times D^{n-k}) \end{tikzcd}$$

(a) **Haeflinger's Step** (a)**:** By definition of $\Phi(D^k_{[1,2]} \times D^{n-k})$, we have $\alpha < 1$ and a factorization of f through $X \times \mathbb{R} \to \Phi(D^k_{[\alpha,2]} \times D^{n-k}) \to \Phi(D^k_{[1,2]} \times D^{n-k})$.

(b) **Haeflinger's Step** (b) **and Geiges' Step** (1)**:** By Microflexibility, we have $\epsilon > 0$ and a lift

$$\begin{tikzcd} & (X \times [0,1]) \times \{0\} \arrow[rr, "f"] \arrow[dl] \arrow[d] & & \Phi(D^k_{[\alpha,2]} \times D^{n-k}) \arrow[d] \\ (X \times [0,1]) \times [0,\epsilon] \arrow[r] \arrow[urr, dashed] & (X \times [0,1]) \times [0,1] \arrow[r, "(f_u, f_{u+v})"'] & \Phi(S^{k-1}_\alpha \times D^{n-k}) \prod \Phi(D^k_{[1,2]} \times D^{n-k}) \arrow[r, "\simeq"] & \Phi((S^{k-1}_\alpha \sqcup D^k_{[1,2]}) \times D^{n-k}) \end{tikzcd}$$

We pick a finite sequence $0 = t_0 < t_1 < \ldots < t_s = 1$ such that for all $0 \leq n < s$, $t_{n+1} - t_n \leq \epsilon$. Denote by $\mu_n \colon (X \times \{t_n\}) \times [t_n, t_{n+1}] \to \Phi(D^k_{[\alpha,2]} \times D^{n-k})$ the restrictions of the lift above. Note that, by construction, the $(\mu_n)_{|S^{k-1}_\alpha}$ are time-independent: we can pick some neighbourhood $S^{k-1}_\alpha \times \{0\} \subset U_\alpha$ on which the μ_n don't depend on t.

(c) **Haeflinger's Step** (c) **and Geiges' Step** (2)**:** By induction, we define lifts

$$\begin{tikzcd} & X \times \{0\} \arrow[r, "g_0"] \arrow[dl] \arrow[d] & \Phi(D^k_2 \times D^{n-k}) \arrow[d] \\ X \times [0,t_n] \arrow[r] \arrow[ur, dashed, "g_n"] & X \times [0,1] \arrow[r, "f"'] & \Phi(D^k_{[\beta,2]} \times D^{n-k}) \end{tikzcd}$$

for some $\alpha < \beta < 1$.

- **Initialization:** We have a pullback square giving us a lift

$$\begin{tikzcd} & X \times \{0\} \arrow[rr, "g_0"] & & \Phi(D^k_{[0,\alpha]} \times D^{n-k}) \arrow[dr] & \\ X \times [0,t_1] \arrow[ur] \arrow[rr, dashed, "g_1"] \arrow[drrr, bend right, "\mu_0"'] & & \Phi(D^k_2 \times D^{n-k}) \arrow[ur] \arrow[dr] & & \Phi(S^{k-1}_\alpha \times D^{n-k}) \\ & & & \Phi(D^k_{[\alpha,2]} \times D^{n-k}) \arrow[ur] & \end{tikzcd}$$

Indeed, the square commutes precisely because $(\mu_0)_{|S^{k-1}_\alpha}$ is time-independent, and hence equal to its value at $t = 0$ being g_0.

- **Induction Step:** Suppose we have already defined g_n. Let U be some neighbourhood of $D^k_{[\alpha,\beta]} \times D^{n-k}$ on which μ_n and f are defined, and which has a nonzero distance to $D^k_{[1,2]} \times D^{n-k}$.

  (1) **Haeflinger's Isotopy and Geiges' Figure** 3.5**:** We pick an isotopy $\psi \colon [0,t_n] \to \mathsf{Emb}(D^k_2 \times D^{n-k})$ such that:

  (i) ψ_t is the identity for $t \leq t_n/3$ and constant for $t \geq 2t_n/3$.

  (ii) ψ is the identity outside of U and on a neighbourhood $S^{k-1}_\beta \times \{0\} \subset U_\beta$ on which f and g_n match (by Induction Hypothesis).

  (iii) ψ_{t_n} maps $S^k_\gamma \times \{0\} \mapsto S^k_\alpha \times \{0\}$ for some $\beta < \gamma < 1$.
  Let $(D^k_2 \times \{0\} \bigcup D^k_{[1,2]}) \times D^{n-k} \subset V$ and $S^{k-1}_\gamma \times \{0\} \subset U_\gamma \subset V$ be neighbourhoods such that U_γ is mapped to U_α by ψ_{t_n}.

  (2) We define $\overline{g}_{n+1} \colon X \times [0,t_{n+1}] \to \Phi(V)$ in a piecewise manner as follows. We first define the following families of sections:

  (I) We let $I \colon X \times [0,t_n] \xrightarrow{g_n} \Phi(D^k_2 \times D^{n-k}) \to \Phi(V_{<\beta} \cup U_\beta)$,

  (II) We let $II \colon X \times [0,t_n] \xrightarrow{(\psi,f)} \mathsf{Emb}(V_{>\beta} \cup U_\beta, V_{>\alpha} \cup U) \times \Phi(V_{>\alpha} \cup U) \to \Phi(V_{>\beta} \cup U_\beta)$.

We can merge these two maps using the sheaf property on the ambiant manifold as is illustrated by the following limit diagram:

$$\begin{tikzcd}
 & & \Phi(D_2^k \times D^{n-k}) \arrow[rr] & & \Phi(V_{<\beta} \cup U_\beta) \arrow[dr] & \\
X \times [0, t_n] \arrow[urr, "g_n"] \arrow[rr, dashed, "\overline{g}_{n+1}"] \arrow[drr, "(\psi, f)"'] & & \Phi(V \cup U_\beta) \arrow[urr] \arrow[drr] \arrow[r, phantom, "\rangle"] & & & \Phi(U_\beta) \\
 & & \mathsf{Emb}(V_{>\beta} \cup U_\beta, V_{>\alpha} \cup U) \times \Phi(V_{>\alpha} \cup U) \arrow[rr] & & \Phi(V_{>\beta} \cup U_\beta) \arrow[ur] &
\end{tikzcd}$$

because on U_β, ψ is the identity, hence the bottom section matches f, which also matches g_n by definition of U_β. We can now define the following maps:

(III) We let $III\colon X \times [t_n, t_{n+1}] \xrightarrow{\mu_n} \Phi(V_{>\alpha} \cup U) \xrightarrow{\psi_{t_n}^*} \Phi(V_{>\gamma} \cup U_\gamma)$,

(IV) We let $IV\colon X \times [t_n, t_{n+1}] \to X \times \{t_n\} \xrightarrow{\overline{g}_{n+1}} \Phi(V) \to \Phi(V_{<\gamma} \cup U_\gamma)$.

Pablo: could be improved: it's strange how this depends on I and II merging

We can merge these two maps using the sheaf property on the ambiant manifold as is illustrated by the following limit diagram:

$$\begin{tikzcd}
 & & \Phi(V_{>\alpha} \cup U) \arrow[r, "\psi_{t_n}^*"] & \Phi(V_{>\gamma} \cup U_\gamma) \arrow[dr] & \\
X \times [t_n, t_{n+1}] \arrow[urr, "g_n"] \arrow[rr, dashed, "\overline{g}_{n+1}"] \arrow[d] & & \Phi(V) \arrow[urr] \arrow[drr] \arrow[r, phantom, "\rangle"] & & \Phi(U_\gamma) \\
X \times \{t_n\} \arrow[rr, "(\overline{g}_{n+1})_{t_n}"'] & & \Phi(V) \arrow[rr] & & \Phi(V_{<\gamma} \cup U_\gamma) \arrow[ur]
\end{tikzcd}$$

because at t_n, ψ maps U_γ inside U_α, on which μ_n is time-independent. By construction, on U_α, μ_n matches f_{t_n}, hence on U_γ, $\psi_{t_n}^* \mu_n$ matches $\psi_{t_n}^* f_{t_n}$.
Using the sheaf property on the parametrizing manifold, we can then merge these two maps to obtain $\overline{g}_{n+1}\colon X \times [0, t_{n+1}] \longrightarrow \Phi(V)$. By construction, the only overlap to check is on $V_{>\gamma} \cup U_\gamma$. The sections mathces because $(\mu_n)_{t_n}$ matches f_{t_n} on $U \supset V_{>\gamma} \cup U_\gamma$.

(3) **Final Isotopy - Squeezing into V:** We pick an isotopy $\xi\colon [0, t_{n+1}] \to \mathsf{Emb}(D_2^k \times D^{n-k})$ such that:
   (i) ξ_0 is the identity,
   (ii) $\xi_t(D_2^k \times D^{n-k}) \subset V$ for $t \geq t_n/4$,
   (iii) ξ_t is the identity near $D_{[1,2]}^k \times D^{n-k}$ for all t.

(4) **Final Construction:** We finally define g_{n+1} using the sheaf property on the parametrizing manifold as follows:

$$\begin{tikzcd}
 & X \times [0, t_n/3] \arrow[rr, "(\xi, g_n)"] \arrow[dr] & & \mathsf{Emb}(D_2^k \times D^{n-k}) \times \Phi(D_2^k \times D^{n-k}) \arrow[dr] & \\
X \times [t_n/4, t_n/3] \arrow[ur] \arrow[dr] & \arrow[r, phantom, "\langle"] & X \times [0, t_{n+1}] \arrow[rr, dashed] & & \Phi(D_2^k \\
 & X \times [t_n/4, t_{n+1}] \arrow[ur] \arrow[rr, "(\xi, \overline{g}_{n+1})"'] & & \mathsf{Emb}(V, D_2^k \times D^{n-k}) \times \Phi(V) \arrow[ur] &
\end{tikzcd}$$

because on $[t_n/4, t_n/3]$, ψ is the identity, so $\overline{g}_{n+1}$ matches g_n, so $\xi^* \overline{g}_{n+1}$ matches $\xi^* g_n$.

(5) **Checking the Lifting Diagram:** Because both ϕ_t and ξ_t are the identity on a neighbourhood of $D_{[1,2]}^k \times D^{n-k}$, and because μ_n agrees with f on $D_{[1,2]}^k \times D^{n-k}$, we have some new $\alpha < \beta' < 1$ such that the desired lifting diagram commutes. ■

Proof of the h-Principle: The directions $(i) \implies (ii) \iff (iii)$ are trivial. We prove here $(ii) \iff (i)$. Firstly, by induction on the index, we show that all the $\Phi(D_{[1,2]}^k \times D^{n-k}) \to \Xi(D_{[1,2]}^k \times D^{n-k})$ are equivalences. Secondly, by induction on handle decomposition, we show that $\Phi(U) \to \Xi(U)$ for all $U \subset M$ open. Find a handle decomposition of U, which is open, by handles of index $< n$, and denote by U_i the union of the first i handles.

Then at each iteration, we have a map of pullback squares

$$\begin{tikzcd}
\Phi(U_{i+1}) \arrow[rr, dashed] \arrow[dr] \arrow[dd, "\circ"'] & & \Xi(U_{i+1}) \arrow[dr] \arrow[dd] & \\
& \Phi(D^k_2 \times D^{n-k}) \arrow[rr, "\simeq"] \arrow[dd] & & \Xi(D^k_2 \times D^{n-k}) \arrow[dd, "\circ"] \\
\Phi(U_i) \arrow[rr] \arrow[dr] & & \Xi(U_i) \arrow[dr] & \\
& \Phi(D^k_{[1,2]} \times D^{n-k}) \arrow[rr, "\simeq"] & & \Xi(D^k_{[1,2]} \times D^{n-k})
\end{tikzcd}$$

where the dotted arrow is forced to be an equivalence by the induction hypothesis. ■

4 Categories and Classifying Spaces

Fix $M = \mathbb{R}^\infty$ and write $\mathcal{D} = \Phi^c(M \times \mathbb{R})$.

Definition 4.1. We let $\mathcal{D}^{\pitchfork}(X)$ be the $\mathrm{Sh}(\mathcal{X})$-internal **poset** of pairs (a, W) where $a\colon X \to \mathbb{R}$ and $W \in \mathcal{D}(X)$ such that

$$\forall x \in X, (W_x \to \mathbb{R}) \pitchfork a$$

Definition 4.2. We let $\mathcal{C}^{\pitchfork}(X)$ be the $\mathrm{Sh}(\mathcal{X})$-internal **category** $\mathcal{C}^{\pitchfork}$ whose:

- collection of objects is the sheaf

$$\mathsf{ob}(\mathcal{C}^{\pitchfork})(X) = \varinjlim_{\epsilon\colon X \to \mathbb{R}} \bigsqcup_{a\colon X \to \mathbb{R}} \mathcal{C}^{\pitchfork}(X, a, a, \epsilon)$$

- collection of morphisms is the sheaf

$$\mathsf{mor}(\mathcal{C}^{\pitchfork})(X) = \varinjlim_{\epsilon\colon X \to \mathbb{R}} \bigsqcup_{a_0 < a_1\colon X \to \mathbb{R}} \mathcal{C}^{\pitchfork}(X, a_0, a_1, \epsilon)$$

  where $\mathcal{C}^{\pitchfork}(X, a_0, a_1, \epsilon)$ is the set of $W \subset X \underline{\times} (a_0 - \epsilon, a_1 + \epsilon) \times M$ such that:

  (a) (smooth family) $W \twoheadrightarrow X$ is a submersion with d-dimensional fibers,

  (b) (compact cobordisms) $W \to X \underline{\times} (a_0 - \epsilon, a_1 + \epsilon)$ is proper,

  (c) (transversality) the corestrictions $W_i \to X \underline{\times} (a_i - \epsilon, a_i + \epsilon)$ for $i \in \{0, 1\}$ are submersions.

We show in this section that the following maps are weak homotopy equivalences:

$$\Omega^{\infty-1} MT(d) \longleftarrow |\mathcal{D}| \longleftarrow B|\mathcal{D}^{\pitchfork}| \longrightarrow B|\mathcal{C}^{\pitchfork}| \longleftarrow B|\mathcal{C}^{\perp}|$$

4.1 First Map: $\Omega^{\infty-1} MT(d) \leftarrow |\mathcal{D}|$

Remark 4.1. For $M = \mathbb{R}^{n+d-1}$, note that we can build an actual map of sheaves $|\mathcal{D}^{\pitchfork}| \to \mathcal{C}^0(-, \Gamma^c(\mathsf{Thom}^{\mathsf{fib}}(\gamma^{\perp}_{b,\mathbb{R}^{n+d-1}}) \to \mathbb{R} \times \mathbb{R}^{n+d-1}))$ by using a "closest point" construction. This can be arranged into a filtered homotopy colimit diagram giving a map of sheaves $|\mathcal{D}| \to \mathcal{C}^0(-, \Omega^{\infty-1} MT(d))$. It's now enough to show this map of sheaves is a weak homotopy equivalence. We **could** do this on pairs (X, A) using a relative version of Phillips Submersion Theorem. Or it's enough to just show surjectivity on π_1 for all base points (or even just for one base point, say the empty manifold, by using the fact that these are $\mathcal{E}_{n+d-1}$-algebras).

This is the **crucial step of the proof**, where the h-Principle is being used. Let X be a compact manifold and take $g\colon (X \times \mathbb{R}^{n+d-1})^+ \to \Omega^{n+d-1}\mathsf{Thom}(\gamma^{\perp}_d)$. By Whitney's Approximation Theorem, we can assume that g is smooth. By the Thom Transversality Theorem, we can assume that $g\colon X \times \mathbb{R}^{n+d-1} \to \mathsf{Thom}(V^{\perp}_d(\mathbb{R}^{n+d}))$ is transverse to the zero section. Let $Y = g^{-1}(\mathsf{Gr}_d(\mathbb{R}^{n+d})) \subset X \times \mathbb{R}^{n+d-1}$ which is a smooth submanifold codimension n.

The **problem** is that $Y \to X$ is **not a submersion** in general. We consider $Y \times \mathbb{R} \to X$ and will use Phillips Submersion Theorem to deform it into a submersion.

We need to construct a vector bundle surjection $T(Y \times \mathbb{R}) \twoheadrightarrow TX$ which will recover the homotopy class of g (at least, by choosing a higher n). Note that we have a caonical identification of the normal bundle of $Y \subset X \times \mathbb{R}^{n+d-1}$ as $g^*(\gamma_d^\perp)$. This gives

$$\begin{cases} g^*(\gamma_d^\perp) \oplus g^*(\gamma_d) \simeq \underline{\mathbb{R}^{n+d}} \\ g^*(\gamma_d^\perp) \oplus TY \simeq TX_{|Y} \oplus \underline{\mathbb{R}^{n+d-1}} \end{cases}$$

From this we get a canonical $\underline{\mathbb{R}^{n+d}} \oplus TY \simeq \underline{\mathbb{R}^{n+d-1}} \oplus TX_{|Y} \oplus g^*(\gamma_d^\perp)$. By **obstruction theory**, we can deform this bundle isomorphism such that it comes from some

$$\underline{\mathbb{R}} \oplus TY \simeq TX_{|Y} \oplus g^*(\gamma_d^\perp)$$

This gives a bundle surjection $T(Y \times \mathbb{R}) \twoheadrightarrow TX$.

Now by **Phillips Submersion Theorem**, we can deform $Y \times \mathbb{R} \xrightarrow{\pi} X$ into a submersion $W = Y \times \mathbb{R} \to X$, by going through bundle surjections. Note that (π, f) is proper where f is the canonical projection $W \to \mathbb{R}$ because Y is compact (because X was compact).

Now, by Whitney's Embedding Theorem, we can find $N >> n$ and an embedding $W \times \mathbb{R} \subset (X \times \mathbb{R}) \times \mathbb{R} \times \mathbb{R}^{N+d-1}$ which lift (π, f) at $t = 0$ and (g, f) at $t = 1$. This shows that we have indeed built a lift.

Relative Version: Similarly, one can use a relative version of Phillips Submersion Theorem to modify the projection away from the preimage of A only. We can also use a relative version of Whitney's Approximation Theorem to ensure that this lifts to an embedding restricting to the constant isotopy on A.

4.2 Second Map: $|\mathcal{D}| \leftarrow B|\mathcal{D}^{\pitchfork}|$

This step uses the **"β"-construction** from the first section and criterion 2.5.

Firstly, we show $\beta\mathcal{D}^{\pitchfork}(X) \to \mathcal{D}(X)$ is surjective. Pick $W \subset_d X \times \mathbb{R} \times M \twoheadrightarrow X$ a submersion with d-dimensional fibers and with $(\pi, f)\colon W \to X \times \mathbb{R}$ proper. For each $x \in X$, choose $a_x \in \mathbb{R}$ a regular value of $f_x\colon W_x \to \mathbb{R}$. Then we can find a neighbourhood $x \in U_x \subset X$ such that a_x is regular of $f_y\colon W_y \to \mathbb{R}$ for all $y \in U_x$. By compactness of X, we can find extract a finite subcover $(U_j)_{\jmath \in J}$ and a list of values $(a_j)_{j \in J}$ such that $(W_{|U_j}, a_j) \in \mathcal{D}^{\pitchfork}(U_j)$. Now picking $a_R = \min\{a_j \mid j \in R\}$, we get $a_R \leq a_S$ and hence a cobordism ϕ_{RS}.

Secondly, we fix $A \subset X$ closed and we show how to extend a lift. This argument can be made relative to a closed subset $A \subset X$, simply by ensuring we pick regular values which are larger than the ones already picked.

4.3 Third Map: $B|\mathcal{D}^{\pitchfork}| \leftarrow B|\mathcal{C}^{\pitchfork}|$

This is the **easiest step of the proof**. We show that the map of sheaves $N_k\mathcal{C}^{\pitchfork} \to N_k\mathcal{D}^{\pitchfork}$ is a weak homotopy equivalence for all $k \geq 0$, using criterion 2.5. It's enough to work on X connected on which $\forall x \in X, a_0(x) < \ldots < a_k(x)$.

Suppose we have $A \subset X$ and $W \subset (X \underline{\times} (a_0 - \epsilon, a_k + \epsilon) \bigcup U \times \mathbb{R}) \times M$ for some open neighbourhood $A \subset U \subset X$ such that:

- restricted to $X \underline{\times} (a_0 - \epsilon, a_k + \epsilon)$, this is an element of $N_k\mathcal{C}^{\pitchfork}(X)$,
- restricted to $U \times \mathbb{R}$, this is an element of $N_k\mathcal{D}^{\pitchfork}(U)$.

We pick a diffeomorphism $\phi\colon X \times \mathbb{R} \simeq X \underline{\times} (a_0 - \epsilon, a_k + \epsilon) \bigcup U \times \mathbb{R}$ which is the identity on a neighbourhood of $X \underline{\times} [a_0 - \epsilon/2, a_k + \epsilon/2] \cup A \times \mathbb{R}$. Then $\phi^* W$ is our desired lift.

4.4 Fourth Map: $B|\mathcal{C}^{\pitchfork}| \to B|\mathcal{C}^{\perp}|$

We show that the map of sheaves $N_k\mathcal{C}^{\pitchfork} \to N_k\mathcal{D}^{\perp}$ is a weak homotopy equivalence for all $k \geq 0$, using criterion 2.5. It's enough to work on X connected on which $\forall x \in X, a_0(x) < \ldots < a_k(x)$.

Pick a bump function $\psi\colon \mathbb{R} \to [0, 1]$ such that:

- $\psi(u) = 0$ if $u \leq 1/3$,
- $\psi(u) = 1$ if $u \geq 2/3$,
- $\psi' > 0$ on $\psi^{-1}((0, 1))$.

Write $\psi(s,u) = \psi(s)\psi(u) + (1-\psi(s))u$. Pick a bump function $\rho\colon X \to [0,1]$ such that $\rho = 0$ on A and $\rho = 1$ on $X \setminus U$ where U is an open neighbourhood of A on which our lift is defined. We define $\Phi\colon X \times \mathbb{R} \times \mathbb{R} \to X$ as follows:

$$\Phi(x,s,u) = a_i(x) + a_{i+1}(x)\psi\left(s\rho(x), \frac{u - a_i(x)}{a_{i+1}(x) - a_i(x)}\right) \qquad \begin{aligned} &\text{for } a_i(x) < u < a_{i+1}(x), \\ &\text{or } i = 0 \text{ and } u < a_1(x), \\ &\text{or } i = k-1 \text{ and } a_{k-1}(x) < u. \end{aligned}$$

which is smooth and transverse to (π, f). We can define $W_\Phi \subset (X \times \mathbb{R}) \times \mathbb{R} \times M$ which is a concordance between the idenity ($s = 0$) and an element in $N_k\mathcal{C}^\perp(X)$ ($s = 1$).

5 The (∞, n)-Category of Cobordisms

In this section, the principles of microflexibility and concordance established in previous sections are elevated to classify the (∞, n)-category of cobordisms. Where the GMTW theorem successfully computes the geometric realization of the 1-fold cobordism category utilizing the h-principle, the natural generalization proposed by Schommer-Pries requires capturing the structure of manifolds with corners structured along an n-dimensional grid.

The prevailing literature often models this structure through an explicit point-set topology on the space of submanifolds, followed by an intricate manipulation of Segal's edgewise subdivision. Here, we depart entirely from the point-set topological approach. Instead, we rigidly adhere to the sheaf-theoretic methodology developed in Sections 1 through 4. By defining the n-fold cobordism category as an n-fold multi-simplicial sheaf over the site $\mathcal{X}$ of smooth manifolds, the geometric operation of "removing cuts" is executed naturally over a test manifold X. The problem is ultimately reduced, via inductive geometric realization, to the microflexible h-principle.

5.1 Definitions and the n-fold Segal Space

Fix integers $d, n \geq 0$, and let M be a fixed, finite-dimensional ambient manifold without boundary. We consider unoriented, unframed submanifolds embedded into the product space $M \times \mathbb{R}^n$.

Definition 5.1. We define an n-fold simplicial sheaf $\mathcal{C}^\perp_{d;n}(M)\colon (\Delta^{\mathsf{op}})^n \to \mathrm{Sh}(\mathcal{X})$ representing the (∞, n)-category of d-dimensional cobordisms. For an object $[\mathbf{k}] = (k_1, \ldots, k_n)\colon (\Delta^{\mathsf{op}})^n$, the collection of sections over a test manifold $X\colon \mathcal{X}$, denoted $\mathcal{C}^\perp_{d;n}(M)_{[\mathbf{k}]}(X)$, consists of pairs $(\mathbf{a}, W)$ satisfying the following structural conditions:

- **The Categorical Grid:** $\mathbf{a} = (\mathbf{a}^1, \ldots, \mathbf{a}^n)$ is a collection of n tuples, where for each $1 \leq i \leq n$, $\mathbf{a}^i = (a^i_0, a^i_1, \ldots, a^i_{k_i})$ is a sequence of smooth maps $a^i_j\colon X \to \mathbb{R}$ such that $a^i_0 < a^i_1 < \cdots < a^i_{k_i}$ everywhere on the parameter space X.
- **The Submanifold:** $W \subset_d X \times \mathbb{R}^n \times M$ is a closed subset such that the projection map $W \twoheadrightarrow X$ is a proper smooth submersion whose fibers are d-dimensional manifolds.
- **Compactness within the Grid:** The projection of the submanifold W into the $\mathbb{R}^n$ factor is strictly bounded by the extremal grid values. Specifically, W is required to be a subset of the localized region $X \underline{\times} \prod_{i=1}^n (a^i_0 - \epsilon, a^i_{k_i} + \epsilon) \times M$ for some strictly positive smooth function $\epsilon\colon X \to (0, \infty)$.
- **Cylindricality (Globularity):** For each coordinate direction $1 \leq i \leq n$ and each grid index $0 \leq j \leq k_i$, the submanifold W must be cylindrical near the hyperplane defined by a^i_j. That is, there exists a neighborhood U of the graph of a^i_j in $X \times \mathbb{R}$ such that W is invariant under translation in the i-th coordinate of $\mathbb{R}^n$ over U.

The face and degeneracy maps of the multi-simplicial structure operate entirely by manipulating the grid $\mathbf{a}$. A face map drops a grid function a^i_j, effectively composing two adjacent cobordisms by forgetting the boundary that separated them. A degeneracy map duplicates a grid function, seamlessly inserting an identity cobordism. The strict cylindricality condition guarantees that these geometric compositions are smooth and well-defined, satisfying the requisite Segal conditions to model a coherent (∞, n)-category.

5.2 The Strategy of Removing Cuts

To compute the total geometric realization $|\mathcal{C}^\perp_{d;n}(M)|$, which evaluates the functor $|-|\colon \mathrm{Sh}(\mathcal{X}) \to \mathbf{Top}$ across all n simplicial directions simultaneously, we must systematically "remove the cuts" defined by the categorical grid.

The grid essentially divides the manifold W into a lattice of n-morphisms. Geometrically, taking the classifying space (geometric realization) of this category with respect to the i-th simplicial direction corresponds to integrating over all possible locations of the cuts a^i_j. Because the specific locations of the cuts do not alter the underlying topological structure of the manifold W, the geometric realization should yield a space equivalent to one where the cuts in the i-th direction have simply been erased.

Let B_n denote the geometric realization functor applied exclusively to the n-th simplicial direction. Our immediate goal is to establish a weak equivalence of $(n-1)$-fold simplicial sheaves:

Once this single-direction reduction is rigorously established, a trivial induction applies. Repeating this process n times systematically removes the entire grid, reducing the complex n-fold categorical realization to the 0-fold classifying space of manifolds embedded in $M \times \mathbb{R}^n$, completely devoid of any categorical cuts:

This 0-fold space is nothing more than the fundamental sheaf of embedded, proper submersions parameterized over X. This is the exact domain where the microflexible h-principle governs.

5.3 Intermediate Categories and Transversality

To prove the equivalence $B_n\left|\mathcal{C}^{\perp}_{d;n}(M)\right| \simeq \left|\mathcal{C}^{\perp}_{d;n-1}(M \times \mathbb{R})\right|$, we isolate the n-th simplicial direction and introduce three intermediate multi-simplicial sheaves. These intermediates mimic the logic of Section 4 by gradually relaxing the stringent cylindricality requirement into a mere transversality requirement, before dropping the cuts entirely.

Definition 5.2. Let $\mathcal{C}^{\pitchfork}_{d;n}(M)$ be the n-fold simplicial sheaf defined identically to $\mathcal{C}^{\perp}_{d;n}(M)$, except the cylindricality condition in the n-th direction is replaced by a standard transversality condition. For each $0 \le j \le k_n$, we require that the mapping a^n_j acts as a regular value for the projection map $W \to \mathbb{R}^{\{n\}}$. The cylindricality conditions for the first $n-1$ directions remain strictly enforced.

Definition 5.3. Let $\mathcal{D}^{\pitchfork}_{d;n}(M)$ be the n-fold simplicial sheaf where the n-th direction represents the nerve of a topological poset. A k_n-simplex over X consists of a manifold W that is bounded and cylindrical in the first $n-1$ directions, but extends unbounded across the entire n-th coordinate $\mathbb{R}$. The simplex data includes a tuple $a^n_0 < a^n_1 < \cdots < a^n_{k_n} : X \to \mathbb{R}$ such that every a^n_j is a regular value of the projection $p\colon W \to \mathbb{R}^{\{n\}}$. Unlike $\mathcal{C}^{\pitchfork}$, the manifold W does not terminate or change geometric structure near the outermost cuts; the cuts merely act as marked regular values along an infinite topological continuum.

Definition 5.4. Let $\mathcal{D}^{\perp}_{d;n}(M)$ denote the $(n-1)$-fold simplicial sheaf formed by completely discarding the n-th grid sequence $\mathbf{a}^n$. A section consists of a manifold $W \subset_d X \times \mathbb{R}^{n-1} \times (M \times \mathbb{R})$ that is cylindrical in the first $n-1$ directions and properly submersed over X. This sheaf is constant with respect to the n-th simplicial index. Note that $\mathcal{D}^{\perp}_{d;n}(M)$ is canonically isomorphic to the reduced category $\mathcal{C}^{\perp}_{d;n-1}(M \times \mathbb{R})$.

We construct a zigzag of natural morphisms of n-fold simplicial sheaves:

We will now prove that, upon applying the geometric realization B_n to the n-th simplicial direction, each of these morphisms induces a weak homotopy equivalence.

5.4 Equivalence of Realizations: Executing the Removal of Cuts

We execute the "removing cuts" argument step-by-step using the criteria for weak homotopy equivalences of sheaves established in Section 2 (specifically Criterion 2.5). We evaluate the spaces over a fixed compact test manifold X and a closed subset $A \subset X$.

5.4.1 Straightening Transversality to Cylindricality

Lemma 5.1

The map $B_n|\mathcal{C}^{\pitchfork}_{d;n}(M)| \to B_n|\mathcal{C}^{\perp}_{d;n}(M)|$ is a weak homotopy equivalence.

Proof: We must demonstrate that any section $u\colon \mathcal{C}^{\pitchfork}_{d;n}(M)(X)$ can be deformed via concordance into a section in $\mathcal{C}^{\perp}_{d;n}(M)(X)$. The disparity between the two sheaves is solely the geometric behavior of W near the hyperplanes a^n_j. In $\mathcal{C}^{\pitchfork}$, W intersects these hyperplanes transversely; in $\mathcal{C}^{\perp}$, W is perfectly cylindrical.

Since transversality is an open condition, we can define a smooth isotopy of the ambient space $X \times \mathbb{R}^n \times M$ that flattens the n-th coordinate of W near the cuts without introducing critical points. Choose a sufficiently small smooth function $\epsilon\colon X \to (0, \infty)$ such that the intervals $(a^n_j - \epsilon, a^n_j + \epsilon)$ are disjoint.

Define a smooth family of diffeomorphisms $\Psi_t \colon X \times \mathbb{R}^n \times M \to X \times \mathbb{R}^n \times M$ parameterized by a time variable t: . The diffeomorphism leaves X, M, and the first $n-1$ coordinates of $\mathbb{R}^n$ entirely invariant. On the n-th coordinate $s \colon \mathbb{R}$, we define Ψ_t utilizing a smooth bump function $\rho(s)$ where $\rho(s) = 0$ for $|s| \leq 1/3$, $\rho(s) = s$ for $|s| \geq 2/3$, and $\rho'(s) > 0$ elsewhere.

We set the coordinate transformation locally near each cut to:

$$\Psi_t(x, s) = (1-t)s + t\left(a_j^n(x) + \epsilon(x)\rho\left(\frac{s - a_j^n(x)}{\epsilon(x)}\right)\right)$$

for s within the ϵ-neighborhood of any cut $a_j^n(x)$, and as the identity function elsewhere.

At $t = 0$, Ψ_0 is the identity mapping. At $t = 1$, Ψ_1 maps a neighborhood of a_j^n constantly to the value a_j^n, effectively "stretching" the transverse intersection into a cylindrical collar of width $\epsilon/3$. Because a_j^n is a strictly regular value, the geometric pullback $W_t = \Psi_t^{-1}(W)$ remains a smooth, proper submersion over X for all t. The resulting family W_t provides a strict, continuous concordance between the transverse cobordism and a fully cylindrical one, fulfilling Criterion 2.5 and proving the equivalence. ■

5.4.2 Gluing the Cobordisms

Lemma 5.2

The inclusion map $B_n|\mathcal{C}^{\pitchfork}_{d;n}(M)| \to B_n|\mathcal{D}^{\pitchfork}_{d;n}(M)|$ is a weak homotopy equivalence.

A k_n-simplex in $\mathcal{C}^{\pitchfork}_{d;n}$ consists of k_n consecutive, compact cobordisms whose boundaries align transversely at the grid cuts a_j^n. Conversely, a k_n-simplex in $\mathcal{D}^{\pitchfork}_{d;n}$ consists of a single, unified manifold W extending indefinitely in the n-th direction, simply marked with transverse regular values $a_0^n < \cdots < a_{k_n}^n$.

The equivalence is proven via a direct deformation retraction over the parameter space X. For any $W \colon \mathcal{D}^{\pitchfork}_{d;n}$, the manifold extends over an unbounded range in $\mathbb{R}^{\{n\}}$. We define a smooth cutoff isotopy that smoothly retracts the infinite tails $(-\infty, a_0^n]$ and where the n-th coordinate of $\mathbb{R}^n$ has been seamlessly absorbed into the ambient background manifold, transitioning the background from M to $M \times \mathbb{R}$.

The (∞, n)-category is structured entirely symmetrically with respect to its n simplicial directions. Therefore, this geometric argument proceeds by simple induction. Applying the geometric realization functor iteratively across all n simplicial directions yields a complete collapse of the grid:

The resulting 0-fold simplicial sheaf $\mathcal{C}^{\perp}_{d;0}(M \times \mathbb{R}^n)$ contains no grid structure, no posets, and no categorical cuts. It is precisely the space of closed, d-dimensional manifolds embedded in $M \times \mathbb{R}^n$ that map down to X as a proper smooth submersion. In the established notation of Section 4, this space is identically $\Phi^c(M \times \mathbb{R}^n)$.

5.5 Final Reduction via the Microflexible h-Principle

We have successfully reduced the geometric realization of the formidable (∞, n)-category of cobordisms down to the bare, geometric sheaf of embedded manifolds $\Phi^c(M \times \mathbb{R}^n)$. The final step is to determine the absolute homotopy type of this space utilizing the microflexible h-principle.

Let the total ambient space be $V = M \times \mathbb{R}^n$. We are interested in evaluating the sheaf $\Phi^c(V)(X)$, which represents the space of d-dimensional submanifolds $W \subset_d X \times V$ that are proper submersions over X.

We strictly analyze the unoriented, unframed geometry of the submanifolds to preserve the general applicability of the proof. For any test point $x \colon X$ and any point on the manifold $w \colon W_x$, the tangent space $T_w W_x$ forms a d-dimensional subspace of the ambient tangent space $T_w V$. This inclusion naturally induces a canonical Gauss map from the submanifold to the Grassmannian bundle:

Because W is smoothly embedded in V, it possesses a well-defined normal bundle $\nu_{W/X}$. Let $m = \dim(M)$. The dimension of the total ambient space V is $m + n$. Therefore, the rank of the normal bundle is exactly the codimension, which is $m + n - d$. Let us define the relative ambient parameter $b = \dim(M) - d = m - d$. The rank of the normal bundle can thus be written as $b + n$.

The normal bundle $\nu_{W/X}$ naturally classifies a continuous mapping into the orthogonal complement of the universal bundle over the Grassmannian. Let γ_d be the tautological d-plane bundle over $\mathsf{Gr}_d(TV)$, and let $\gamma_d^{\perp}$ be its orthogonal complement of rank $b+n$. The normal bundle $\nu_{W/X}$ is canonically isomorphic to the pullback $\tau^*_{W/X}(\gamma_d^{\perp})$.

According to the microflexible h-principle (derived directly from Phillips's Submersion Theorem in Section 3), the sheaf of actual geometric submersions $\Phi^c(V)$ is weakly homotopy equivalent to the sheaf of *formal* submersions. A formal submersion drops the rigid geometric integrability condition of the tangent bundle. It replaces the strict geometric embedding with purely bundle-theoretic data: a continuous map into V accompanied by an injective bundle map over it.

By applying the Pontryagin-Thom collapse construction fiberwise over the ambient space V, the data of a formal submanifold equipped with a normal bundle $\gamma_d^\perp$ translates directly and bijectively into a continuous map into the Thom space of that bundle.

Specifically, we form the fiberwise Thom space $\mathsf{Thom}^{\mathsf{fib}}(\gamma_d^\perp) \to V$. The fiber over a point $v\colon V$ is the classic Thom space of the vector bundle $\gamma_d^\perp$ restricted to the Grassmannian $\mathsf{Gr}_d(T_vV)$. The Pontryagin-Thom construction dictates that for a test space X, the formal bundle data over $X \times V$ is equivalent to a compactly supported section of this fiberwise Thom bundle.

Therefore, the h-principle furnishes a natural weak homotopy equivalence:

This expression gives the space of compactly supported sections over the full ambient space $M \times \mathbb{R}^n$. However, we can refine this. Because the space $\mathbb{R}^n$ is contractible, the bundle topology over $M \times \mathbb{R}^n$ retracts smoothly onto the topology over the base manifold M. The compactly supported sections extending over the $\mathbb{R}^n$ factor correspond topologically to the n-fold loop space operation Ω^n.

Integrating out the $\mathbb{R}^n$ directions, we recover the final, simplified target space over the base manifold M:

where $b = \dim(M) - d$.

This concludes the proof. The geometric realization of the unoriented (∞, n)-category of cobordisms embedded in M is wholly determined by the space of compactly supported sections of the fiberwise Thom fibration over M. The rigorous implementation of the multi-simplicial h-principle completely circumvents the necessity for complex point-set topological approximations, leveraging the robustness of the sheaf-theoretic β-construction to natively and elegantly eliminate the categorical grid cuts.

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